
Midterm Practice Problems

Econ 502: Advanced Microeconomics

Problem 1: Monopoly Pricing

A monopolist faces the inverse demand curve $P = 200 - 2Q$ and has a constant marginal cost of $MC = 40$ with no fixed costs.

- Derive the marginal revenue function. Find the profit-maximizing quantity Q_m^* , price P_m^* , and the monopolist's profit.
- What would the equilibrium price and quantity be under perfect competition (i.e., $P = MC$)? Calculate total surplus (consumer surplus + producer surplus) under perfect competition.
- Calculate total surplus under monopoly and the deadweight loss from monopoly pricing.
- Compute the Lerner Index at the monopoly equilibrium. Verify your answer using the inverse elasticity rule.

Problem 2: Third-Degree Price Discrimination

A streaming service has two types of subscribers: adults (A) and students (S). The demand in each market is:

- Adults: $Q_A = 100 - P_A$
- Students: $Q_S = 60 - P_S$

The marginal cost of serving each additional subscriber is constant at $c = 20$.

- If the firm can price discriminate between the two markets, find the optimal price, quantity, and profit in each market.
 - If the firm must charge a single (uniform) price to all subscribers, find the profit-maximizing price, total quantity, and total profit.
 - Compute the price elasticity of demand and the Lerner index in each market under price discrimination. Which market gets the higher markup? Explain the intuition.
 - Compare total consumer surplus and total welfare (CS + profit) under the two pricing regimes. Is price discrimination welfare-improving here?
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Problem 3: Cournot Duopoly

Two firms compete by simultaneously choosing quantities. Market demand is $P = 150 - Q$, where $Q = q_1 + q_2$. Both firms have constant marginal cost $c = 30$ and no fixed costs.

- Derive each firm's best response (reaction) function.
- Find the Cournot-Nash equilibrium: each firm's output, the market price, and per-firm profits.
- Suppose the two firms collude to maximize joint profits and split output equally. Find the collusive quantity per firm, price, and per-firm profit. If firm 2 sticks to the collusive output, what would firm 1 optimally produce? Would firm 1 want to deviate from the agreement?
- Fill in the following table comparing three market structures (you already have the Cournot and collusion results; compute the competitive outcome where $P = MC$):

	Price	Total Output	Per-Firm Profit
Competition			
Cournot			
Collusion (Monopoly)			

Problem 4: Bertrand Competition

Two firms produce a homogeneous good and compete by simultaneously choosing prices. Consumers buy from the firm with the lower price; if both charge the same price, they split the market equally. Both firms have constant marginal cost c .

- Suppose both firms set prices $p_1 = p_2 = c$. Show that this is a Nash equilibrium by verifying that neither firm can profitably deviate.
- Explain why any pair of prices with $p_1 = p_2 > c$ cannot be a Nash equilibrium.
- Now suppose costs are asymmetric: firm 1 has $MC_1 = 10$ and firm 2 has $MC_2 = 20$. The market demand is $Q = 100 - P$. What is the Nash equilibrium? What is firm 1's profit? (*Hint: Think about what happens if firm 1 prices just below firm 2's cost.*)
- The result in parts (a) and (b) is known as the *Bertrand paradox*: just two firms are enough to eliminate all market power. Give two assumptions of the Bertrand model that, if relaxed, would allow firms to earn positive profits in equilibrium.

Problem 5: Utility Maximization and Welfare

A consumer has utility function $U(x, y) = xy$ and income $I = 100$. The initial prices are $p_x = 1$ and $p_y = 1$.

- Derive the Marshallian demand functions $x^*(p_x, p_y, I)$ and $y^*(p_x, p_y, I)$ by solving the utility maximization problem. Find the optimal bundle and utility level at the initial prices.

- b) Suppose the price of x rises to $p'_x = 4$ (with $p_y = 1$ and $I = 100$ unchanged). Find the new optimal bundle. Using the Slutsky decomposition, calculate the substitution effect (holding money income fixed so the consumer can just afford the original bundle) and the income effect separately.
- c) Derive the expenditure function $E(p_x, p_y, \bar{U})$ by solving the expenditure minimization problem. Use it to compute the Compensating Variation (CV) for the price increase in part (b). Interpret the CV in words.
- d) Using the Marshallian demand functions, determine whether x is a normal or inferior good, and whether x and y are gross substitutes or complements. Now use the expenditure function from part (c) to find the Hicksian demand for x and determine whether x and y are net (Hicksian) substitutes or complements. Reconcile the two results.

Problem 6: Production, Cost Minimization, and Firm Supply

A competitive firm produces output using capital K and labor L with the production function:

$$Q = K^{1/3}L^{1/3}$$

Factor prices are $r = w = 1$.

- a) What are the returns to scale of this production function? Does this mean that if the firm doubles all inputs, output more than doubles, exactly doubles, or less than doubles? How does this affect the long-run average cost curve?
- b) Set up the cost minimization problem and derive the conditional factor demands $K^*(r, w, Q)$ and $L^*(r, w, Q)$. Use these to derive the total cost function $C(r, w, Q)$. Evaluate C at $r = w = 1$.
- c) Derive the average cost $AC(Q)$ and marginal cost $MC(Q)$ at $r = w = 1$. Show that $MC > AC$ for all $Q > 0$. What does this imply about the shape of the average cost curve, and is it consistent with the returns to scale you found in part (a)?
- d) The output price is $p = 6$. Find the profit-maximizing output Q^* . Calculate total revenue, total cost, and profit at Q^* . Does the firm earn positive, zero, or negative economic profit? Is this consistent with a long-run competitive equilibrium?

Problem 7: Competitive Equilibrium and Tax Incidence

The market for a good has demand and supply curves:

$$Q^D = 240 - 2P \quad Q^S = 3P - 60$$

- a) Find the competitive equilibrium price P^* and quantity Q^* . Calculate consumer surplus (CS), producer surplus (PS), and total welfare.
- b) The government levies a specific tax of $t = 25$ per unit on producers. Find the new equilibrium consumer price P^D , producer price $P^S = P^D - t$, and quantity Q_t .

- c) Calculate the price elasticities of demand and supply at the pre-tax equilibrium. Using the tax incidence formula, determine what fraction of the tax burden falls on consumers versus producers. Verify by comparing $P^D - P^*$ and $P^* - P^S$ directly.
- d) Compute the tax revenue, the deadweight loss from the tax, and the changes in consumer surplus and producer surplus. Verify that $\Delta CS + \Delta PS + \text{Tax Revenue} + \text{DWL} = 0$.

Problem 8: Exchange Economy and General Equilibrium

Consider an exchange economy with two consumers (A and B) and two goods (x and y). Their endowments are:

$$\omega_A = (8, 2), \quad \omega_B = (2, 8)$$

Their utility functions are $U_A(x_A, y_A) = x_A y_A$ and $U_B(x_B, y_B) = x_B y_B$.

- a) Compute each consumer's MRS at the endowment point. Is the initial allocation Pareto efficient? Explain intuitively which good each consumer would want to trade away and why.
- b) Derive the contract curve (the set of all Pareto efficient allocations) in the Edgeworth box. Express it as a relationship between x_A and y_A , and describe its geometry.
- c) Let $p = p_x/p_y$ be the price ratio. Find the competitive equilibrium: derive each consumer's Walrasian demands, apply market clearing to find p^* , and state the equilibrium allocation. Verify that the equilibrium lies on the contract curve.
- d) State the First and Second Welfare Theorems. A social planner wants to implement the Pareto efficient allocation $(x_A, y_A) = (3, 3)$ instead of the laissez-faire equilibrium. Identify the lump-sum transfer (in terms of a reallocation of endowments) that, combined with competitive trading at the equilibrium price ratio, supports this allocation.

Problem 9 (Conceptual): The Welfare Theorems

- a) State the First Welfare Theorem and the assumptions it requires. Give two real-world reasons why a competitive equilibrium might fail to be Pareto efficient.
- b) State the Second Welfare Theorem. Why is this theorem important for policy — what does it tell us about the relationship between efficiency and equity?
- c) A politician says: “*The First Welfare Theorem proves that free markets produce the best outcomes for society, so the government should never intervene.*” Do you agree with this interpretation? Explain why or why not.

Problem 10 (Conceptual): Taxes, Efficiency, and Incidence

- a) A senator proposes taxing producers rather than consumers, arguing this shields consumers from the burden. Using the theory of tax incidence, evaluate this claim. Under what (extreme) condition would it actually be true that producers bear the entire burden?

- b) A per-unit tax and a lump-sum tax both raise the same revenue. Which causes a larger deadweight loss? Explain the economic intuition — what is it about a per-unit tax that makes it distortionary?
 - c) Consider a market where demand is perfectly inelastic. If the government imposes a per-unit tax, who bears the burden? What is the deadweight loss? Now answer the same questions if supply is perfectly elastic instead.
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Problem 11 (Conceptual): Monopoly, Elasticity, and Market Power

- a) A profit-maximizing monopolist never operates on the inelastic portion of the demand curve. Explain why, using the relationship between marginal revenue and elasticity.
 - b) Using the inverse elasticity rule $L = 1/|\varepsilon_D|$, explain why a monopolist facing more elastic demand charges a lower markup. What happens to the Lerner Index as demand becomes perfectly elastic, and what does this imply about the monopolist's behavior?
 - c) Consider Cournot competition with n identical firms. What happens to the equilibrium price, total output, and per-firm profit as n increases? What market structure does Cournot converge to in the limit? How does this compare to the Bertrand model with just $n = 2$ firms?
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Problem 12 (Conceptual): Consumer Theory

- a) What is the difference between ordinal and cardinal utility? Why do economists generally work with ordinal utility? Give an example of a result in consumer theory that relies only on ordinal properties.
 - b) Using the Slutsky equation, explain why a Giffen good must be an inferior good. Can a normal good ever be Giffen? Can an inferior good fail to be Giffen?
 - c) Under what conditions are the Compensating Variation (CV), Equivalent Variation (EV), and the change in Marshallian Consumer Surplus (ΔCS) all equal to each other? Why do they generally differ?
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Problem 13 (Conceptual): Production and Market Structure

- a) Explain the relationship between returns to scale and the shape of the long-run average cost curve. Why might an industry with strong increasing returns to scale tend toward a monopoly (a *natural monopoly*)?
- b) A competitive firm sets $P = MC$, while a monopolist sets $MR = MC$. Both are profit-maximizing. Why does the competitive firm *ignore* the effect of its output on price while the monopolist does not? What key structural assumption explains the difference?
- c) In the long run with free entry and exit, competitive firms earn zero economic profit. Does this mean they earn zero accounting profit? Explain the difference and why firms still choose to operate.

Problem 14: Quasilinear Utility

A consumer has utility $U(x, y) = 2\sqrt{x} + y$, where y is a numeraire good with $p_y = 1$. The price of x is p and income is I (assume I is large enough for an interior solution throughout).

- Derive the Marshallian demand function for x . Show that demand for x does not depend on income. What does this tell you about the income elasticity of demand for x ?
 - Show that for this utility function, the Marshallian and Hicksian demands for x coincide. What does this imply about the substitution and income effects of a price change?
 - Suppose p rises from 1 to 4. Calculate the change in Marshallian consumer surplus, the Compensating Variation, and the Equivalent Variation. What do you notice?
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Problem 15: Two-Part Tariff

A monopolist serves a single consumer whose demand for a good is $Q = 20 - P$. The monopolist has $MC = 4$ with no fixed costs.

- Under uniform (per-unit) pricing, find the profit-maximizing price, quantity, and profit. Calculate consumer surplus and deadweight loss.
 - Now the monopolist uses a *two-part tariff*: a fixed entry fee T plus a per-unit price p . The consumer buys only if their consumer surplus (net of T) is non-negative. Find the optimal two-part tariff (T^*, p^*) and the monopolist's total profit.
 - Compare total welfare (CS + profit) under uniform pricing and the two-part tariff. Which regime is more efficient? Who gains and who loses relative to uniform pricing?
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Problem 16: Short-Run Supply and Long-Run Equilibrium

A competitive firm has the short-run total cost function $C(Q) = 50 + 2Q^2$, where 50 is a sunk fixed cost.

- Derive $MC(Q)$, $ATC(Q)$, and $AVC(Q)$. Find the output level that minimizes ATC and the corresponding minimum ATC .
 - What is the firm's short-run supply curve? At what price does the firm shut down in the short run? At what price does it break even (zero economic profit)?
 - If the market price is $p = 40$, find the firm's profit-maximizing output and profit.
 - In the long run with free entry/exit of identical firms, what is the equilibrium price? If market demand is $Q^D = 200 - 5P$, how many firms operate in the long-run equilibrium?
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Problem 17: Stackelberg Competition

Two firms compete in quantities with inverse demand $P = 100 - Q$, where $Q = q_1 + q_2$. Both have $MC = 10$ and no fixed costs. Firm 1 is the Stackelberg leader: it chooses q_1 first, and firm 2 (the follower) observes q_1 before choosing q_2 .

- Derive firm 2's best response function $q_2^*(q_1)$.
- Write down firm 1's profit as a function of q_1 only (substituting in firm 2's best response). Find the Stackelberg equilibrium: q_1^* , q_2^* , market price, and each firm's profit.
- Compare the Stackelberg outcome to the Cournot equilibrium (where both firms move simultaneously). Who benefits from being the leader? Is total output higher or lower than in Cournot?
- Compare total welfare (CS + total profit) under Stackelberg and Cournot. Which is closer to the competitive outcome?

Problem 18: Subsidy and Welfare

The market for electric vehicles has:

$$Q^D = 500 - 5P \quad Q^S = 5P - 100$$

- Find the equilibrium price and quantity. Calculate CS and PS.
- The government provides a per-unit subsidy of $s = 20$ to producers. Find the new consumer price P^D , producer price $P^S = P^D + s$, and quantity.
- Calculate the government's total cost of the subsidy. What is the change in CS, the change in PS, and the deadweight loss?
- Who benefits more from the subsidy — consumers or producers? How does the relative benefit relate to the elasticities of demand and supply?

Problem 19: Comparative Statics with Elasticities

In a competitive market, demand has a constant price elasticity $\varepsilon_D = -2$ and supply has a constant price elasticity $\varepsilon_S = 3$. The initial equilibrium is $P^* = 10$ and $Q^* = 100$.

- Write down the demand and supply functions consistent with these elasticities and the initial equilibrium. (*Hint: use the constant-elasticity forms $Q^D = A \cdot P^{\varepsilon_D}$ and $Q^S = B \cdot P^{\varepsilon_S}$.*)
- Suppose a shift in tastes increases demand by 10% (i.e., A increases by 10%). Using the elasticity formula for comparative statics, find the percentage change in the equilibrium price and quantity. (*Hint: $\hat{P} = \frac{\hat{A}}{\varepsilon_S - \varepsilon_D}$ and $\hat{Q} = \varepsilon_S \cdot \hat{P}$, where hats denote percentage changes.*)
- Would the price increase be larger or smaller if supply were more elastic? Explain the intuition.

Problem 20: Efficiency Conditions in a Two-Good Economy

An economy produces two goods, x and y , using labor. The production functions are $x = 2L_x$ and $y = L_y$, with a total labor endowment of $\bar{L} = 100$.

- Derive the Production Possibilities Frontier (PPF). What is the Marginal Rate of Transformation (MRT)?
 - A representative consumer has utility $U(x, y) = xy$. Find the efficient allocation: the quantities of x and y that maximize utility subject to the PPF.
 - Suppose the economy is competitive. Let the wage be w and the prices of x and y be p_x and p_y . Using zero-profit conditions in each industry, find the relationship between p_x , p_y , and w . Show that the resulting price ratio p_x/p_y equals the MRT , and that the competitive equilibrium replicates the efficient allocation.
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Problem 21: Cournot with Asymmetric Costs

Two firms compete in quantities with inverse demand $P = 120 - Q$, where $Q = q_1 + q_2$. Firm 1 has $MC_1 = 20$ and firm 2 has $MC_2 = 40$.

- Derive each firm's best response function.
 - Find the Nash equilibrium quantities, the market price, and each firm's profit.
 - Compute the Lerner index for each firm. Which firm has more market power? Explain.
 - How does total output and welfare compare to the symmetric case where both firms have $MC = 30$? What drives the difference?
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Problem 22: Income and Substitution Effects with Perfect Complements

A consumer has preferences $U(x, y) = \min\{x, 2y\}$ and income $I = 120$. The prices are $p_x = 2$ and $p_y = 3$.

- Find the optimal consumption bundle.
 - Now suppose p_x rises to 4. Find the new optimal bundle and the change in demand for x .
 - Decompose the change in demand for x into a substitution effect and an income effect. What is special about the substitution effect for perfect complements? Explain the intuition.
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Problem 23: Connecting Consumer and Producer Theory

A competitive industry has N identical consumers, each with demand $q^d = 50 - P$, and M identical firms, each with supply $q^s = P - 10$ (for $P \geq 10$).

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- a) Derive the market demand and market supply curves. Find the equilibrium price and quantity as functions of N and M .
 - b) Let $N = 100$ and $M = 50$. Find the equilibrium price and quantity. Calculate per-consumer surplus, per-firm profit, and total welfare.
 - c) In the long run, firms enter until profit is driven to zero. If each firm has $C(q) = 10q + \frac{1}{2}q^2 + 50$ (where 50 is a fixed cost), find the long-run equilibrium price, per-firm output, and the number of firms when $N = 100$.
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